

Di Yang

Applied Machine Learning & Multimodal Generation AI | Ph.D. Researcher | Summer 2027 Internship

Williamsburg, VA | +1 757-332-9351 | dyang06@wm.edu | Google Scholar | LinkedIn

Research Profile

Ph.D. researcher working on multimodal learning, generative models, and reinforcement learning. Experience includes 5B policy fine-tuning, few-step model distillation, GPU kernel development, and training and evaluation pipelines. Research spans efficient generation, sequential decision-making, and learning under partial observability.

Education

William & Mary

Ph.D. Student in Data Science | Advisor: Dr. Yanhai Xiong

Williamsburg, VA
Sep 2024 - Expected Dec 2027

National University of Singapore

M.Sc. in Electrical Engineering | Advisor: Dr. John S. Ho

Singapore
Jan 2021 - Jan 2022

University of Electronic Science and Technology of China

B.Eng. in Microelectronic Science and Engineering

Chengdu, China
Sep 2016 - Jul 2020

Selected Publications

Peer-reviewed

- **D. Yang** et al. "LENS: Learning to Navigate with Active Search for Partially Observable MAPF in Unknown Environments." *TMLR*, 2026.
- **D. Yang**, Y. Xiong. "UP-Bench: A Benchmark for Underwater Path Planning Algorithms." *KDD 2025 Benchmark Track (Oral)*.

Manuscripts under review

- **D. Yang**, Z. Li, Y. Xiong, Y. Wang. "Channel Match: Calibrating Shape Latents for Few-Step 3D Generation." Under review at **AAAI 2027**.
- H. Jiang, Y. Li, **D. Yang***, Y. Xiong, H. Chen, Y. He. "VTV-FM: Flow Matching through Variational Terminal-Velocity Closure." Under review at **NeurIPS 2026**. (*Co-first author)
- Y. Li, **D. Yang***, H. Chen, Y. Xiong. "HyCO: A Hybrid Neural Solver for Combinatorial Optimization." Under review at **NeurIPS 2026**. (*Co-first author)

Selected Research Experience

SparcAI Inc.

Research Intern | Advisor: Dr. Yufei Wang

Jan 2026 - Present
Remote

Multimodal Policy Learning for Vision-Language-Action Manipulation

- Integrating robot-centric point maps inspired by *See Like a Robot* into a **5B VLA policy** to incorporate metric spatial information into manipulation.
- Built OmniGibson/BEHAVIOR training and evaluation pipelines aligning RGB-D, robot state, language, and actions; developed a **failure-case-driven data augmentation pipeline**.
- Configured module-specific learning rates for downstream fine-tuning, using a lower rate for the pretrained VLM backbone than for the action DiT and projector.

Channel Match: Efficient Few-Step Generative Modeling

- Distilled a many-step native 3D generator into a **two-step** student using continuous self-consistency training, achieving **4.5x faster** shape sampling.
- Implemented a custom **JVP FlashAttention kernel** for forward-mode differentiation over sparse structured 3D latents.
- Identified channel-wise latent drift affecting downstream texture synthesis; proposed a training-free gated calibration without model retraining.

William & Mary

Ph.D. Researcher | Advisor: Dr. Yanhai Xiong

Sep 2024 - Present
Williamsburg, VA

HyCO: Hybrid Reinforcement Learning and Diffusion for Optimization

- Designed an RL-to-diffusion handover using policy entropy and model disagreement to combine sequential construction with generative solution completion.
- Reduced TSP-1000 optimality gap from **15.4% to 4.6%** over the RL backbone; matched a 16-sample diffusion baseline on TSP-100 with one guided completion.

Additional Research

William & Mary

Ph.D. Researcher | Advisor: Dr. Yanhai Xiong

Sep 2024 - Present

Williamsburg, VA

LENS & UP-Bench: Generalization and Evaluation for Decision-Making

- Proposed **LENS** (TMLR), separating macro-topological reasoning from local collision avoidance for partially observable multi-agent pathfinding in unknown environments.
- Matched the OOD generalization of an 85M-parameter MAPF foundation model using **less than 0.2%** of its training data.
- Designed and open-sourced **UP-Bench** (KDD 2025 Benchmark Track, **Oral**), a 3D underwater simulator with fluid dynamics and dynamic obstacles; benchmarked classical planners and SAC agents.

VTV-FM: Second-Order Flow Matching

- Co-developed phase-space flow matching with cubic Hermite trajectories and a closed-form minimum-acceleration variational closure.
- Evaluated across synthetic systems, PDE fields, and CIFAR-10, achieving **FID 2.99** versus 3.45-3.73 for compared baselines.

SparcAI Inc.

Research Intern | Advisor: Dr. Yufei Wang

Jan 2026 - Present

Remote

Sparse Scene VAE for Geometry-Aware Representation Learning

- Developing a sparse-voxel VAE that encodes posed RGB-D observations into a compact 3D latent and decodes **2D Gaussian surfels** for scene reconstruction.
- Built differentiable rendering supervision using photometric, depth, and normal losses.

Engineering Experience

Cambricon Technologies Co., Ltd.

Digital Logic Engineer

Aug 2022 - Jul 2024

- Built functional verification pipelines for large-scale AI accelerator chips under timing and scalability constraints.
- Owned system-level performance verification for low-latency data flow and error-handling paths.

Technical Skills

Machine learning: multimodal policy fine-tuning, reinforcement learning (SAC, PPO), imitation learning, diffusion, flow matching, consistency distillation, representation learning

Implementation: Python, PyTorch, Hugging Face, Triton, distributed training (DDP/multi-GPU), sparse convolution, gsplat, Linux

Data & evaluation: RGB-D/video pipelines, failure-case data augmentation, simulation benchmarks, OmniGibson, BEHAVIOR-1K, HoloOcean

Honors

Summer Research Fellowship, William & Mary (2025, 2026)

Scholarship for Model Student | University of Electronic Science and Technology of China(2018, 2019)